AI-Assisted Data Analytics Final Project

Project Overview

The objective of this project is to apply the complete data analytics and machine learning workflow to a real-world dataset. You will select an appropriate dataset, prepare it for analysis, explore and visualize the data, build predictive machine learning models, and communicate your findings through an interactive dashboard and a video presentation.

This project is designed to demonstrate your ability to integrate data analytics, machine learning, visualization, and Generative AI to solve a real-world business problem.

AI Usage Policy

AI tools (such as ChatGPT, Microsoft Copilot, Gemini, or similar tools) may be used to assist your work. However, AI should support your learning—not replace it. You are responsible for understanding every step of your project and verifying the accuracy of all AI-generated content.

Project Workflow

Part 1 — Dataset Selection

Select a dataset that satisfies **all** of the following requirements.

General Requirements

The dataset must:

- Be a **structured tabular dataset** (organized into rows and columns).
- Be related to a real-world domain such as:
    - Sales
    - Finance
    - Human Resources (HR)
    - Marketing
    - Healthcare
    - Education
    - Customer Analytics
    - E-commerce
- Contain **at least 1,000 records (rows)**.

- Contain **at least 8 variables (columns)**.

Data Characteristics

The dataset should:

- Include both **numerical** and **categorical** variables.
- Contain meaningful relationships between variables.
- Include a **clearly defined target variable** suitable for machine learning.
- Allow feature engineering and predictive modeling.

Data Quality

The dataset should contain some real-world imperfections, such as:

- Missing values.
- Duplicate records.
- Incorrect or inconsistent values.
- Incorrect data types.
- Outliers or unusual observations.

These issues provide an opportunity to demonstrate data cleaning and preprocessing skills.

Analysis Potential

The dataset should allow you to:

- Perform exploratory data analysis.
- Build predictive machine learning models.
- Generate business insights.
- Create an interactive dashboard.

Dataset Justification

In your report, explain:

- Why you selected the dataset.
- The business problem you aim to solve.
- Why the target variable is appropriate for prediction.
- Why the dataset is suitable for this project.

Part 2 — Data Preparation

Using Python (Pandas and NumPy) and, where appropriate, Microsoft Excel, prepare the dataset for analysis.

Where applicable, your preprocessing should include:

- Handling missing values.
- Removing duplicate records.
- Correcting inconsistent values.
- Fixing incorrect data types.
- Standardizing formatting.
- Detecting and treating outliers.
- Encoding categorical variables.
- Feature scaling (when appropriate).
- Feature engineering.
- AI-assisted recommendations for improving data quality.

Document all major cleaning steps in your report.

Part 3 — Exploratory Data Analysis (EDA)

Analyze the dataset to understand its characteristics and discover meaningful insights.

Your analysis may include:

- Summary statistics.
- Distribution of variables.
- Category comparisons.
- Trend analysis.
- Grouping and aggregation.
- Relationships between variables.
- Outlier detection.
- Business observations and insights.

Your goal is to answer meaningful questions about the data rather than simply creating charts.

Part 4 — Predictive Modeling

Develop machine learning models to predict the selected target variable.

Your project should include:

- Selection of the target variable.
- Feature selection.

- Train-test split.
- Model training.
- Model evaluation.
- Comparison of model performance.

Implement and compare the following models where applicable:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Linear Regression (for regression problems)

Evaluate your models using appropriate evaluation metrics.

Examples include:

- Classification
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion Matrix
- Regression
  - MAE
  - MSE
  - RMSE
  - R^2 Score

Explain why the best-performing model was selected.

Part 5 — AI-Generated Insights

Use Generative AI to assist in interpreting your results.

Your AI-generated summary should include:

- Explanation of important trends.
- Interpretation of model performance.
- Business implications.
- Actionable recommendations.

All AI-generated content must be reviewed and verified before submission.

Part 6 — Interactive Dashboard

Using **Power BI** or **Tableau**, create an interactive dashboard that communicates the results of your project.

Your dashboard should include:

- KPIs and summary metrics.
- Interactive filters and slicers.
- Meaningful visualizations.
- Predictive insights where appropriate.
- Clear titles and labels.
- Professional layout and formatting.

KPI Requirements

Your dashboard must include **at least one Key Performance Indicator (KPI).**

For each KPI, explain:

- What the KPI measures.
- The target value used.
- How the target was determined.
- Why the target is appropriate.
- Whether the KPI met or fell short of the target.

Note: *A KPI is a metric that measures progress toward an important business objective. For this project, every KPI **must** be compared against a clearly defined target or benchmark. The target provides context, allowing you to determine whether performance is meeting expectations, exceeding them, or falling short.*

Part 7 — Face-Cam Presentation

Record a **5–10 minute** presentation explaining your project from start to finish.

Your presentation should include:

- A brief introduction to the dataset and business problem.
- An overview of the data cleaning and preparation process.
- Key findings from the exploratory data analysis (EDA).
- The machine learning workflow, including the models implemented.
- A comparison of the models and explanation of the final model selected.
- A demonstration of the interactive dashboard.
- A discussion of the AI-generated insights and recommendations.
- Final conclusions and key takeaways.

Presentation Requirements

The presentation must include:

- **Screen recording** showing your notebook and dashboard.
- **Face-cam video** throughout the presentation.
- Clear audio.
- A duration of **5 to 10 minutes**.

Deliverables

Submit **all** of the following files.

1. Dataset Description (PDF)

Provide a brief description of the selected dataset.

The document should include:

- Dataset title.
- Dataset source.
- Business domain.
- Number of rows and columns.
- Description of each variable.
- Why you selected the dataset.
- Business problem or analytical objective.
- Why the dataset satisfies the project requirements.

2. Cleaned Dataset

Submit the cleaned dataset used throughout the project.

The workbook should include:

- The cleaned dataset.
- All cleaning reflected in the final data.
- Appropriate formatting.
- No unnecessary worksheets or temporary calculations.

Accepted formats:

- Excel (.xlsx)
- CSV (.csv)

3. Jupyter Notebook (.ipynb)

The notebook should include:

- Data loading.
- Data cleaning.
- Feature engineering.
- Exploratory Data Analysis.
- Machine learning models.
- Model evaluation.
- AI-assisted analysis.
- Clear comments and explanations.

4. Interactive Dashboard

Submit the project file containing your dashboard.

The dashboard should include:

- Meaningful visualizations.
- At least one KPI measured against a target.
- Slicers and interactive filters.
- Clear titles and labels.
- Consistent formatting and layout.
- A professional appearance.
- Predictive insights.
- Insights that help answer the business questions.

Accepted formats:

- Power BI (.pbix)
- Tableau (.twb /.twbx)

4. Project Report (PDF)

Prepare a report (approximately 3–6 pages) summarizing your work.

Your report should include the following sections.

1. Dataset Overview

- Dataset description.
- Business domain.
- Business problem.
- Target variable.
- Project objectives.

2. Data Preparation

Describe:

- Data quality issues.
- Cleaning process.
- Feature engineering.
- AI-assisted preprocessing.

3. Exploratory Data Analysis

Summarize:

- Trends.
- Relationships.
- Outliers.
- Important observations.

4. Machine Learning

Explain:

- Models implemented.
- Evaluation metrics.
- Model comparison.
- Best-performing model.
- Justification for model selection.

5. Dashboard Summary

Describe:

- Dashboard design.
- KPIs.
- Visualizations.
- Predictive insights.

6. KPI Explanation

For each KPI explain:

- KPI definition.
- Target value.
- Target justification.
- Performance compared to the target.

7. AI-Generated Insights

Summarize:

- AI-generated observations.
- Business recommendations.
- Validation of AI outputs.

8. Final Conclusions

Summarize the overall findings and recommendations.

Submission Checklist

Before submitting, ensure your submission contains all of the following:

- Dataset Description (PDF)
- Cleaned Dataset (.xlsx or .csv)
- Jupyter Notebook (.ipynb)
- Interactive Dashboard (.pbix or .twbx)
- Project Report (PDF)
- Face-Cam Walkthrough Video

Incomplete submissions may receive a grade deduction.